Group-Based Trajectory Modeling using NLMIXED

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1. Overview

This metadata provides a structured description of the datasets, variables, and macros used in the Group-Based Trajectory Modeling using NLMIXED SAS code. The code is designed to estimate latent trajectory classes for two longitudinal outcomes (T1 and T2) based on repeated measures data.

2. Datasets Overview

Dataset Name	Description
`base_file_srs`	Main dataset containing longitudinal measures of two outcomes (T1, T2) for individuals.
`parameter`	Stores estimated parameter values from the NLMIXED model.
`data_pred`	Merged dataset with individual-level predictions and model parameters.
`pred_membership_y`	Dataset containing posterior class memberships for each individual.
`avg_membership`	Aggregated dataset with average posterior class memberships.
`wide`	Transposed dataset containing latent class estimates for visualization.
`data_plot`	Merged dataset for plotting predicted vs. observed outcomes.

3. Variable Definition

Key Identifiers

Variable Name	Type	Description
`ID`	CHAR	This variable identifies the subject ID, and it is saved in the output file for future linkage between the predicted group and external variables. ID=id.

class_all	CHAR	List of latent class labels used for trajectory modeling.
INDEP	NUM	Independent variables. e.g. INDEP=Time_1-Time_8
VAR	NUM	Dependent variables. The number of dependent variables and the number of independent variables should match, e.g. VAR=Y_1-Y_8.
LC	NUM	Number of latent classes.
Starting	NUM	The initial values for the GBTM fitting iteration. For example,%starting_value_alpha(class=3) assigns a starting value of 0 to the proportion estimation for class 3 (indicating equal probability in class assignment). If not explicitly provided, the default starting value for the model parameters is set to 1. e.g. starting= %starting_value_alpha(class=3) sigma_=10.
Order	NUM	Polynomial order for the model.
Equal_sigma	NUM	(T=equal variance, F= unequal variance) This specifies whether the program assumes that the error term in the model has a consistent variance across the distribution.
MAX	NUM	Upper bounds for truncated normal distribution

Time-Related Variables

Variable Name	Type	Description
T	NUM	Number of repeated time points per individual (e.g., 12 quarters).
X[T]	Array	Time variable used as a predictor in trajectory models.

Outcome Variables

Variable Name	Type	Description
Y1[T]	Array	Outcome 1
Y2[T]	Array	Outcome 2

Model Parameters

Variable Name	Type	Description
alpha0_[class]	NUM	Intercept for each trajectory class.
beta_[class]_[order]	NUM	Polynomial coefficients modeling outcome trends.

sigma_[outcome]_[class]	NUM	Variance parameter for each class.
mu_[class]_[outcome][T]	Array	Predicted mean outcome for each class at time T.

Class Membership Variables

Variable Name	Type	Description
pie_[class]	NUM	Probability of belonging to each latent trajectory class.
pinumer_[class]	NUM	Numerator in posterior probability computation.
pideno	NUM	Denominator in posterior probability computation.
post_[class]	NUM	Posterior probability of individual belonging to a class.

Likelihood & Residuals

Variable Name	Type	Description
e_[outcome]_[class]	NUM	Residual (Observed - Predicted) for each class.
ll_latclass	NUM	Log-likelihood value for each latent class.

4. Macro Function Definition

Macro Name	Purpose
%starting_value_alpha	Initializes starting values for alpha parameters across classes.
%starting_value_beta_sigma	Sets initial values for beta coefficients and sigma (variance).
%bounds_alpha	Defines parameter constraints to stabilize model estimation.
%bounds_sigma	Ensures sigma (variance) parameters stay positive.
%initiation_universal	Creates universal arrays for time, outcome, and class membership.
%initiation	Initializes class-specific and outcome-specific arrays.
%model	Constructs polynomial trajectory equations for each class.
%residual	Computes residuals for each class by subtracting observed values from predictions.
%float_control	Applies outlier control by capping extreme residuals.
%Prob_cnorm	Calculates probability density function values for normal distribution.
%Class_Membership	Computes class membership probabilities.
%LogLike	Computes log-likelihood for single outcome models.
%LogLike_multi	Computes log-likelihood for multi-trajectory models.
%Posterior	Computes individual posterior probabilities for class assignment.

`%plot_prep`	Prepares data for visualizing observed vs. predicted trajectories.
`%nlmixed_1`	Runs single-outcome trajectory modeling using `PROC NLMIXED`.
`%nlmixed_MultiTraj`	Runs joint trajectory modeling for two outcomes.

5. Expected Output

Output Name	Description
`nlm_fix_T1_[class]`	Parameter estimates for single-outcome trajectory models (T1).
`nlm_fix_T2_[class]`	Parameter estimates for single-outcome trajectory models (T2).
`nlm_fix_T1_T2_[class]`	Parameter estimates for joint trajectory models (T1 & T2).
`avg_membership`	Averaged posterior class membership probabilities.
`pred_membership_y`	Individual-level class assignment probabilities.
`data_plot`	Dataset for plotting predicted vs. observed values.

Notes & Assumptions

- This model assumes **no missing values** in outcome variables (Y1, Y2). Missing data must be imputed before running this analysis.
- The class labels (A, B, C, D, etc.) are predefined, and any changes must be reflected in the macro variable %let class_all.
- If equal_sigma=T, a **single variance** (sigma) is estimated across classes. Otherwise, variance is estimated per class.
- PROC NLMIXED requires well-defined starting values. If the model fails to converge, modify %starting_value_beta_sigma.